$\therefore g^{-1}(\{-1,0,1\}) = \{x \mid -1 \le x < 2\}$

c) $g^{-1}(\{x \mid 0 < x < 1\})$. By defn. of floor function, we know that it always returns an integer. $\therefore g^{-1}$ does not exist. □

46) Let f be a function from A to B. Let S and T be subsets of B. Show that a) $f^{-1}(S \cup T) = f^{-1}(S) \cup f^{-1}(T)$

Soln: Given, $S \subseteq B$ and $T \subseteq B$.

$f^{-1}(S \cup T) = \{x \mid x \in A \text{ and } f(x) = y, \text{ where } y \in S \cup T\}$

Let, $a \in f^{-1}(S \cup T)$. $\therefore f(a) \in S \cup T$. By Defn. 1 from Sec-2.2, $f(a) \in S \vee f(a) \in T$. $\therefore a \in f^{-1}(S) \vee a \in f^{-1}(T)$. By Defn. 1 from Sec-2.2 $a \in f^{-1}(S) \cup f^{-1}(T)$. By Defn. 3 from Sec-2.1, $f^{-1}(S \cup T) \subseteq f^{-1}(S) \cup f^{-1}(T)$

Let, $x \in f^{-1}(S) \cup f^{-1}(T)$. By Defn. 1 from Sec-2.2, $x \in f^{-1}(S) \vee x \in f^{-1}(T)$. $\therefore f(x) \in S \vee f(x) \in T$. By Defn. 1 from Sec-2.2, $f(x) \in S \cup T$. $\therefore x \in f^{-1}(S \cup T)$. By Defn. 3 from Sec-2.1, $f^{-1}(S) \cup f^{-1}(T) \subseteq f^{-1}(S \cup T)$. $\therefore f^{-1}(S \cup T) = f^{-1}(S) \cup f^{-1}(T)$ □

b) $f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T)$

Soln: Let, $x \in f^{-1}(S \cap T)$. By Defn. of inverse image, $f(x) \in S \cap T$. By Defn. 2 from Sec-2.2, $f(x) \in S \wedge f(x) \in T$. By Defn. of inverse image, $x \in f^{-1}(S) \wedge x \in f^{-1}(T)$. By Defn. 2 from Sec-2.2, $x \in f^{-1}(S) \cap f^{-1}(T)$. By Defn. 3 from Sec-2.1, $f^{-1}(S \cap T) \subseteq f^{-1}(S) \cap f^{-1}(T)$

Let, $x \in f^{-1}(S) \cap f^{-1}(T)$. By Defn. of intersection from Sec-2.2, $x \in f^{-1}(S) \wedge x \in f^{-1}(T)$. By Defn. of inverse image, $f(x) \in S \wedge f(x) \in T$. By Defn. 2 from Sec-2.2, $f(x) \in S \cap T$. By Defn. of inverse image, $x \in f^{-1}(S \cap T)$. By Defn. 3 from Sec-2.1, $f^{-1}(S) \cap f^{-1}(T) \subseteq f^{-1}(S \cap T)$. $f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T)$ □